Teacher's Signature:___________
Date: ______________________

United International **University**

Name: ____ ID: ______________

Section: ______ Batch: _____________ Date: ________________________

Experiment No. 06

Name of the Experiment: Determination of the radius of curvature of a plano-convex lens by Newton's rings method.

Introduction

In this experiment the physical property of *interference of light* will be used to determine the radius of curvature of a plano-convex lens. The interference fringe system here is a pattern of concentric circles (See figure 3), the diameter of which you will measure with a travelling microscope (which has a Vernier scale). If a clean convex lens is placed on a clean glass slide (optically flat) and viewed in monochromatic light, a series of rings may be seen around the point of contact between the lens and the slide. These rings are known as *Newton's rings* (See figure 3.) and they arise from the interference of light reflected from the glass surfaces at the air film between the lens and the slide. The experimental set-up is shown in figure 1.

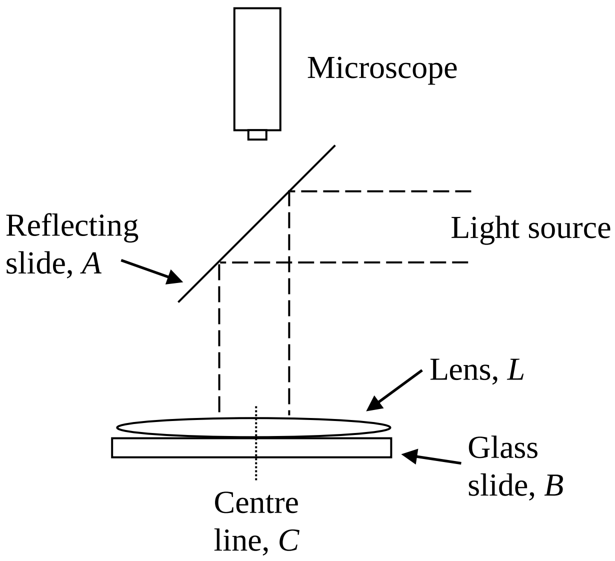

Figure 1: Apparatus

Theory:

Let R be the radius of curvature of the lower surface of the lens. Let r_n be the radius of the n^{th} dark ring, to be measured with the microscope.

Let the corresponding thickness of the air gap at the point P be t. (See figure 4.)

The path difference between the beams reflected at Q and P is approximately $2t$ (for vertical viewing, at small radius r).

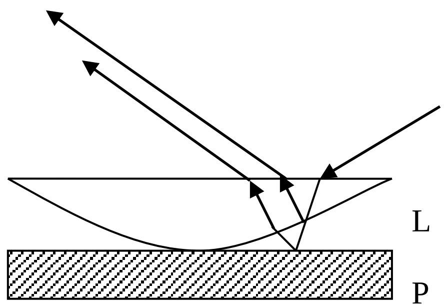

Figure 2

From geometry (refer to figure 5),

$$R^2 = r_n^2 + (R-t)^2$$

$$or, R^2 = r_n^2 + R^2(1-\frac{t}{R})^2$$

$$or, R^2 = r_n^2 + R^2(1-2\frac{t}{R}+\frac{t^2}{R^2})$$

Since $\left(\frac{t}{r}\right)^2$ is very small, we can neglect the term

$$\therefore R^2 = r_n^2 + R^2 - 2Rt$$

$$or, r_n^2 = 2Rt$$

But, we know, for dark fringes, $2t = n\lambda$

Thus, $r_n = \sqrt{nR\lambda}$

So, for n^{th} and $(n+p)^{th}$ ring we can write

$$D_n^2 = 4nR\lambda$$

$$and, D_{n+p}^2 = 4(n+p)R\lambda$$

Hence, we have,

$$D_{n+p}^2 - D_n^2 = 4pR\lambda$$

$$or, R = \frac{D_{n+p}^2 - D_n^2}{4p\lambda}$$

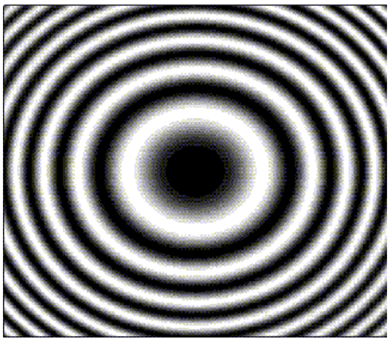

Figure 3: Newton's Rings

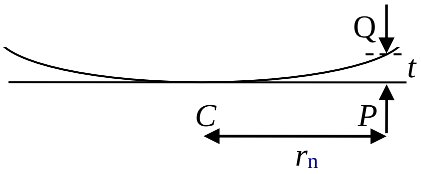

Figure 4

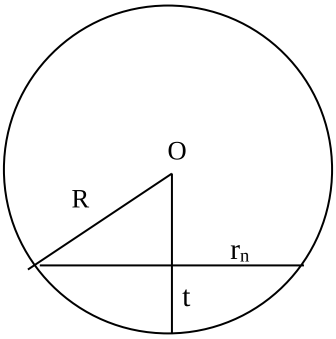

Figure 5: Determining D_n

Taking data to measure the radius of curvature of the plano-convex lens

In this part of the experiment you will measure the diameter of ten rings using the Vernier scale on the traveling microscope, and then use this data to determine the radius of curvature of the convex lens. .

The light from the sodium lamp is partially reflected downwards by a glass slide A. The beams reflected from the lens, L and the glass slide B goes through the slide A to the microscope.

Look for the interference rings with the naked eye – it is easiest to spot these from a height and changing your viewing angle. You may need to maneuver the reflecting slide until you can clearly view the rings.

Focus the microscope on the fringes and align the cross-hair tangential to the central dark spot.

Measure the diameters of at least ten dark rings by setting the cross-hair on one side of a series of rings, reading the positions and then moving the microscope to the other side of the corresponding rings.

You could start measuring the position of the 12^{th} ring, proceeding to the 11^{th}, 10^{th}, etc. and then moving across to the other side of the central ring until you have measured the 12^{th} ring again.

Read the Vernier scale precisely.

Draw a graph of D_n^2 against the ring number (order number of the fringe) n.

Hence calculate the value of R from the graph.

Apparatus:

1) Traveling microscope 2) Plano-convex lens 3) Sodium lamp

Experimental Data:

Least Count:

Ring no. (n)	Readings of the microscope								Diameter of ring, D = L~R (cm.)	D^2 (cm^2)
	Left Side (L)				Right Side (R)					
	L.S.R. x (cm.)	C.S.R.	Value of C.S.R. y (cm.)	Total, x+y (cm.)	L.S.R. x (cm.)	C.S.R.	Value of C.S.R. y (cm.)	Total, x+y (cm.)		

Calculation:

Result:

The radius of curvature of the lower surface of the given lens, *R* =

Discussions:

Q: Explain why do you observe the dark and bright colored fringes? Why is the central spot dark?

Q: If we now use transmitted light instead of reflected light, will there be any change in the center spot? Explain.

Q: As like this experiment, thin film interference is occurred in the thin film of soapy water in a bubble, as a result it looks multicolored. Explain why and how a soap bubble looks multi colored.

Q: In Newton's ring experiment why the glass plate (in our given apparatus it is circular) is place at 45^0?

Q: Why are the fringes circular?

Q: Why should a lens of large radius of curvature is used in this experiment?